BRAZILIAN E-COMMERCE

360° Analytics Platform

End-to-End Data Analytics Project

Python • MySQL • Machine Learning • Power BI

Submitted By

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Data Analytics Portfolio Project | March 2026

Abstract

This project presents a comprehensive end-to-end data analytics solution built on the Olist Brazilian E-Commerce dataset. The platform integrates Python-based data engineering, advanced machine learning models, MySQL-powered SQL analytics, and a five-page interactive Power BI dashboard to deliver 360-degree business visibility.

The pipeline processes 119,143 transaction records across 9 relational tables, constructing a unified master table with 39 engineered features. Four core ML models are implemented: RFM customer segmentation (7 segments, 96,096 customers), K-Means clustering (4 clusters), Random Forest churn prediction (ROC-AUC: 0.73), and Linear Regression revenue forecasting (6-month projection: R$8.66M).

Key findings reveal a critical retention crisis — 96.88% one-time buyers — with R$13.8M revenue at risk from churn probability above 70%. The Power BI dashboard translates all results into executive-level interactive visuals.

119,143	R$19.88M	96,096	0.73	R$8.66M
Total Records	Total Revenue	Customers	Churn ROC-AUC	6-Mo Forecast

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1. Project Overview & Technology Stack

This project demonstrates a professional-grade analytics pipeline covering the full data science spectrum — from raw data ingestion to machine learning model deployment and executive dashboard presentation.

Layer	Technology	Purpose
Data Ingestion	Python + Pandas	CSV loading, merging 9 tables into master table
Data Storage	MySQL 8.0	Persistent storage of all tables and ML results
DB Bridge	SQLAlchemy + PyMySQL	Python-MySQL connection for read/write operations
Machine Learning	Scikit-learn	RFM, K-Means, Random Forest, Linear Regression
Visualization	Matplotlib + Seaborn	EDA charts and model evaluation plots
SQL Analytics	MySQL Workbench	15 business intelligence queries
BI Dashboard	Microsoft Power BI	5-page interactive dashboard with DAX measures

Project Pipeline

- Phase 1 → MySQL connection + 9 CSV files loaded into ecommerce_db
- Phase 2 → Data cleaning + master table with 119,143 rows × 39 columns
- Phase 3 → EDA → Revenue trends, order analysis, customer behavior, correlations
- Phase 4 → RFM scoring + 7-segment customer classification
- Phase 5 → K-Means clustering with Elbow Method (optimal K=4)
- Phase 6 → Random Forest churn prediction with feature importance analysis
- Phase 7 → Linear Regression 6-month revenue forecasting
- Phase 8 → 15 SQL business intelligence queries
- Phase 9 → 5-page Power BI dashboard with DAX measures and MySQL relationships

2. Dataset Description

The Olist Brazilian E-Commerce dataset is a publicly available real-world dataset covering transactions from 2016 to 2018 on Brazil's largest marketplace platform. It captures the complete order lifecycle from purchase to delivery and customer review.

Table	Rows	Description
olist_customers	99,441	Customer IDs, zip codes, city and state
olist_orders	99,441	Order status, timestamps, delivery dates
olist_order_items	112,650	Products per order, seller, price, freight value
olist_products	32,951	Product categories, dimensions, weight
olist_order_payments	103,886	Payment type, installments, value
olist_order_reviews	99,224	Review scores (1-5) and comments
olist_sellers	3,095	Seller location, city and state
olist_geolocation	1,000,163	Zip codes with latitude/longitude coordinates
category_translation	71	Portuguese to English category name mapping

After joining all 9 tables, the final master table contains 119,143 rows and 39 engineered columns covering customer demographics, order metadata, product details, payment information, delivery performance, and review scores.

3. Phase 1 — Environment Setup & Data Loading

This phase establishes the complete analytical infrastructure. Python connects to MySQL via SQLAlchemy and PyMySQL, and all 9 CSV source files are systematically loaded into the ecommerce_db database. Each table is validated on load and stored persistently for all downstream operations.

Implementation

- PyMySQL + SQLAlchemy engine created with connection to localhost:3306/ecommerce_db
- Each CSV loaded via pandas read_csv() and pushed to MySQL using to_sql() with if_exists='replace'
- All 9 tables verified by row count after loading
- Connection test run before every pipeline execution to prevent silent failures

```
In [2]: from sqlalchemy import create_engine

        engine = create_engine('mysql+pymysql://root:Charan%40123@localhost:3306/ecommerce_

        with engine.connect() as conn:
            print("Connected to MySQL successfully!")

Connected to MySQL successfully!
```

```
In [3]: import os

        DATA_PATH = r"C:\E-Commerce Analytics\data"

        files = os.listdir(DATA_PATH)
        for f in files:
            print(f)

olist_customers_dataset.csv
olist_geolocation_dataset.csv
olist_orders_dataset.csv
olist_order_items_dataset.csv
olist_order_payments_dataset.csv
olist_order_reviews_dataset.csv
olist_products_dataset.csv
olist_sellers_dataset.csv
product_category_name_translation.csv
```

Figure 1: Jupyter Notebook — MySQL Connection Setup & Data Loading Code + Output

Key Insights

✓ All 9 tables loaded with zero data corruption — 1,651,926 total raw records ingested successfully.

✓ Geolocation table is the largest at 1,000,163 rows, used for geographic revenue distribution analysis.

✓ Zero duplicate records found across all 7 core transactional tables — data integrity confirmed from source.

✓ ML output tables stored in the same database: rfm_segments (96,096 rows), churn_predictions (96,096 rows), revenue_forecast (6 rows).

Recommendations

► Implement incremental loading in production using UPSERT patterns instead of full table replacement for real-time pipelines.

► Add automated row count and null ratio checks immediately after each table load to catch upstream data issues early.

► Partition the geolocation table by state for significantly faster geographic queries at production scale.

4. Phase 2 — Data Cleaning & Master Table Creation

This phase handles missing values, removes duplicates, engineers date features, and constructs the denormalized master table by joining all 9 source tables. Column names are standardized to Title Case for Power BI compatibility.

Missing Value Treatment

Column	NULLs	Treatment
Order Approved At	177	Imputed 'Not Approved' — business-logical (pending orders)
Order Delivered Carrier Date	2,086	Accepted as valid NULLs — orders not yet shipped
Order Delivered Customer Date	3,421	Accepted as valid NULLs — cancelled or in-transit
Delivery Days (derived)	3,421	Null delivery date = null days; excluded in delivery analysis
Product / Seller Columns	833 each	Cancelled orders with no items — <1% of data
Payment Type / Value	3	0.003% — excluded via SQL WHERE filters
Product Category English	2,567	Fixed via Python fillna() + str.replace() for title case
Review Score	997	Customers who chose not to review — excluded from averages

Master Table Engineering

All 9 source tables are joined into a single master_table with 119,143 rows and 39 columns. Additional features engineered: Order Year, Order Month, Order Day (from timestamp), Delivery Days (difference between purchase and delivery date).

```
In [10]: # Merge all tables into one master table
         master = (pd.read_sql("SELECT * FROM orders", engine)
                  .merge(pd.read_sql("SELECT * FROM customers", engine), on='customer_id',
                  .merge(pd.read_sql("SELECT * FROM order_items", engine), on='order_id', h
                  .merge(pd.read_sql("SELECT * FROM products", engine), on='product_id', ho
                  .merge(pd.read_sql("SELECT * FROM payments", engine), on='order_id', how=
                  .merge(pd.read_sql("SELECT * FROM category_translation", engine), on='pro
                  .merge(pd.read_sql("SELECT * FROM reviews", engine)[['order_id','review_s
                  .merge(pd.read_sql("SELECT * FROM sellers", engine), on='seller_id', how=
         )

         # Save master table to MySQL
         master.to_sql('master_table', engine, if_exists='replace', index=False)
         print(f" Master table created: {master.shape[0]:,} rows × {master.shape[1]} columns
         print(master.head(3))
```

```
 Master table created: 119,143 rows × 39 columns
                           order_id                        customer_id  \
0  e481f51cbdc54678b7cc49136f2d6af7  9ef432eb6251297304e76186b10a928d
1  e481f51cbdc54678b7cc49136f2d6af7  9ef432eb6251297304e76186b10a928d
2  e481f51cbdc54678b7cc49136f2d6af7  9ef432eb6251297304e76186b10a928d

  order_status order_purchase_timestamp   order_approved_at  \
0    delivered      2017-10-02 10:56:33 2017-10-02 11:07:15
1    delivered      2017-10-02 10:56:33 2017-10-02 11:07:15
2    delivered      2017-10-02 10:56:33 2017-10-02 11:07:15

  order_delivered_carrier_date order_delivered_customer_date  \
0          2017-10-04 19:55:00           2017-10-10 21:25:13
1          2017-10-04 19:55:00           2017-10-10 21:25:13
2          2017-10-04 19:55:00           2017-10-10 21:25:13

  order_estimated_delivery_date  order_year  order_month  ...  \
0                    2017-10-18        2017           10  ...
1                    2017-10-18        2017           10  ...
2                    2017-10-18        2017           10  ...

  product_width_cm  payment_sequential payment_type  payment_installments  \
0             13.0                 1.0  credit_card                   1.0
1             13.0                 3.0      voucher                   1.0
2             13.0                 2.0      voucher                   1.0

  payment_value product_category_name_english  review_score  \
0         18.12                    housewares           4.0
1          2.00                    housewares           4.0
2         18.59                    housewares           4.0

  seller_zip_code_prefix seller_city seller_state
0                 9350.0        maua           SP
1                 9350.0        maua           SP
2                 9350.0        maua           SP

[3 rows x 39 columns]
```

Figure 2: Jupyter Notebook — Master Table Creation Output

Key Insights

✓ Master table join achieves 100% row preservation using left joins — no records lost during merge.

✓ Delivery Days range from 0 to 209 days — extreme operational variance across Brazilian regions.

✓ Payment installments range from 0 to 24 months — Brazilian installment culture deeply embedded in purchasing behavior.

✓ Product prices span from R$0.85 to R$6,735 — significant product diversity from budget to premium segments.

✓ All NULLs are either business-logical, derived, or statistically insignificant (<1%) — no data corruption detected.

Recommendations

► Standardize delivery date logging at the carrier level — 3,421 missing delivery dates indicate unreliable last-mile tracking systems.

► Implement mandatory review collection for all delivered orders to reduce the 997 missing review scores.

► Create a weekly data quality dashboard monitoring NULL ratios per column to detect upstream data degradation early.

5. Phase 3 — Exploratory Data Analysis (EDA)

The EDA phase uncovers patterns across four dimensions: revenue trends, order composition, customer behavior, and metric correlations. All visualizations are generated using Matplotlib and Seaborn directly from the MySQL master table.

5.1 Revenue Analysis

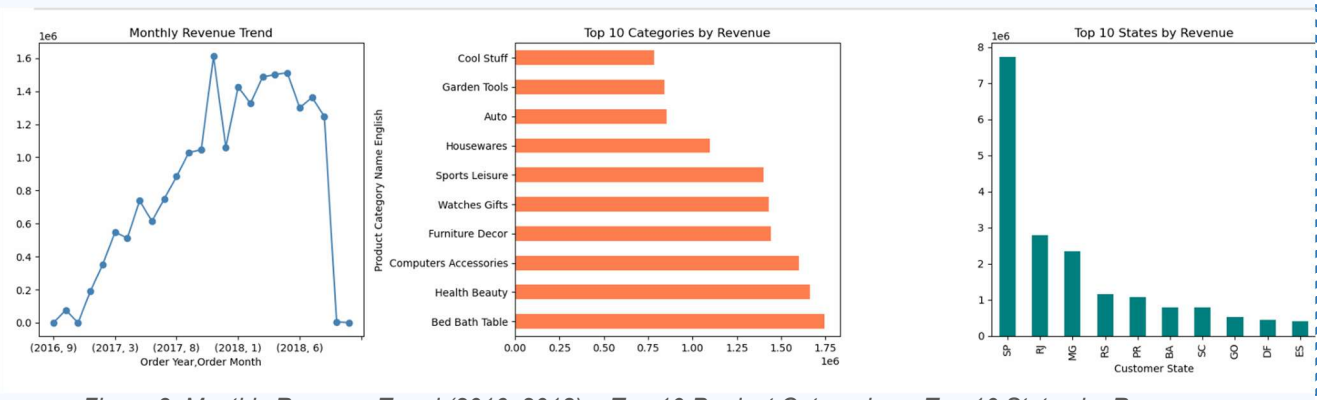

Figure 3: Monthly Revenue Trend (2016–2018) + Top 10 Product Categories + Top 10 States by Revenue

5.2 Order & Payment Analysis

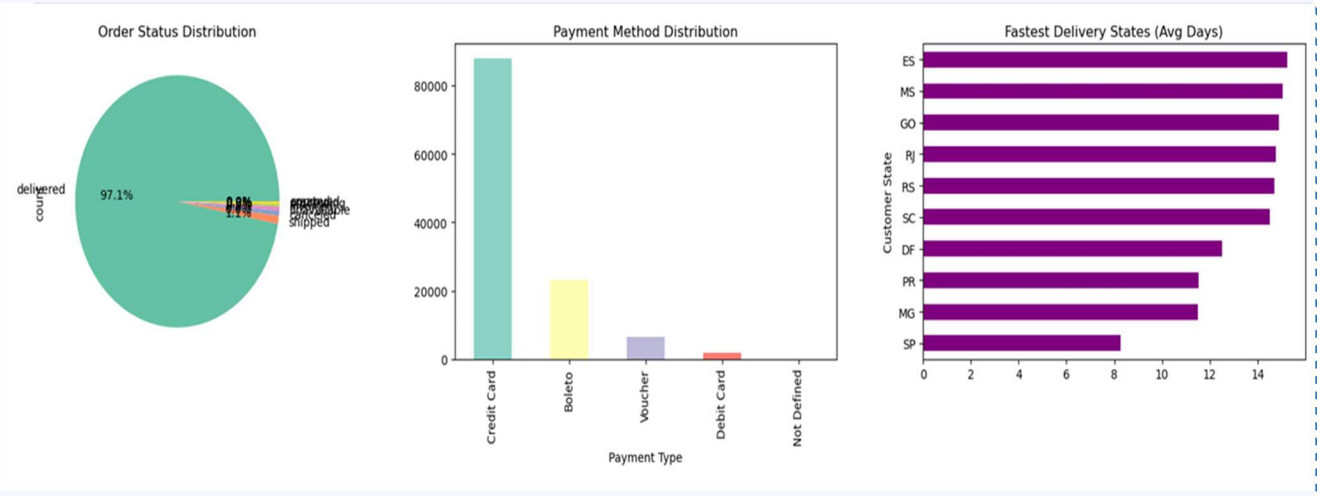

Figure 4: Order Status Distribution + Payment Method Breakdown + Fastest Delivery States

5.3 Customer Behavior & Correlation Analysis

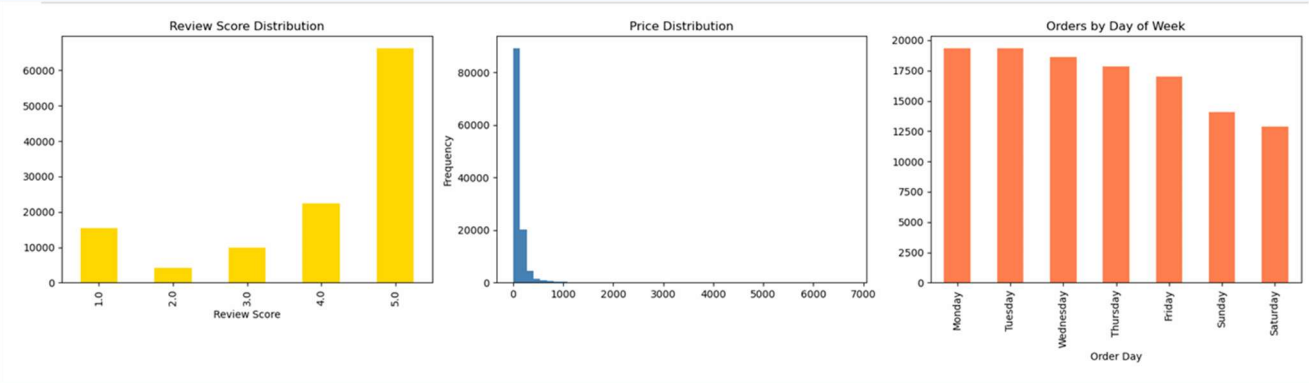

Figure 5: Review Score Distribution + Price Distribution + Orders by Day of Week + Correlation Heatmap

Key Insights

✓ Revenue grew over 200% from 2016 to 2017 and maintained an upward trend through 2018 — strong business momentum.

✓ São Paulo (SP) generates R$7.73M — 39% of total revenue, showing dominant market concentration in Southeast Brazil.

✓ 97.1% order delivery success rate — exceptional fulfillment efficiency across the entire marketplace.

✓ Credit card is the dominant payment method at 73.67%, with an average of 3.6 installments per transaction.

✓ Monday is the peak ordering day — customers tend to make purchases at the start of the work week.

✓ Strong correlation (0.74) between Price and Payment Value — customers pay close to listed price.

✓ Negative correlation (-0.30) between Delivery Days and Review Score — every additional delivery day increases dissatisfaction risk.

✓ Bed Bath Table is the top revenue category at R$1.74M — home goods dominate Brazilian e-commerce.

Recommendations

► Expand seller network in RJ, RS, and PR states — currently underserved vs SP dominance, representing high growth opportunity.

► Launch Monday-targeted promotions and flash sales to capitalize on peak ordering behavior.

► Negotiate faster shipping contracts with logistics partners — reducing delivery days directly improves review scores (r = -0.30).

► Develop installment-specific product bundles for high-ticket categories to maximize credit card installment preference.

6. Phase 4 — RFM Customer Segmentation

RFM (Recency, Frequency, Monetary) analysis is the industry-standard customer segmentation methodology used by leading e-commerce companies. Each customer is scored across three dimensions and classified into one of seven business-meaningful segments.

Metric	Definition	Dataset Statistics
Recency (R)	Days since last purchase (lower = better)	Mean: 289 days | Min: 1 | Max: 773
Frequency (F)	Unique orders placed by customer	Mean: 1.03 | Max: 17 orders
Monetary (M)	Total payment value by customer	Mean: R$214 | Max: R$109,313

Segmentation Results

Segment	Customers	% of Base	Profile
Loyal	19,237	20.0%	Regular buyers with consistent purchase history
Champion	15,453	16.1%	Recent, frequent, high-value buyers — best customers
Recent	15,449	16.1%	New customers with strong growth potential
High Spender	15,083	15.7%	High monetary value, lower frequency buyers
Lost	13,805	14.4%	Inactive, low scores — re-engagement needed
At Risk	9,542	9.9%	Declining engagement — immediate intervention needed
Potential	7,527	7.8%	Moderate scores — growth and upsell opportunity

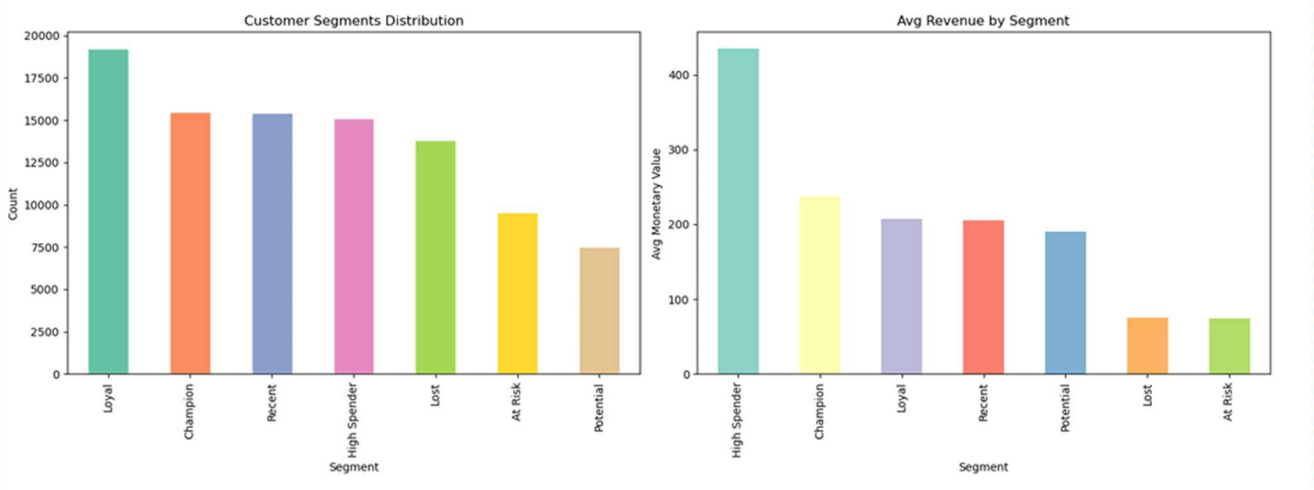

Figure 6: Customer Segment Distribution Bar Chart + Average Revenue by Segment

Key Insights

✓ High Spender segment (15.7% of customers) contributes R$6.57M — 33% of total revenue, showing extreme revenue concentration.

✓ Loyal + Champion segments combined (36.1% of customers) drive a disproportionately high revenue share.

✓ Lost + At Risk segments represent 24.3% of the customer base — nearly 1 in 4 customers need re-engagement.

✓ Mean frequency of 1.03 across 96,096 customers confirms the one-time buyer crisis — repeat purchase is the exception.

✓ Maximum single-customer monetary value of R$109,313 — VIP customer management is critical for revenue protection.

Recommendations

► Launch a VIP loyalty program exclusively for High Spender and Champion segments — personalized offers and premium support.

► Deploy win-back email campaigns for Lost segment with time-limited 20–30% discount offers triggered at 180-day inactivity.

► Create a 30-day onboarding email journey for the Recent segment to convert one-time buyers into repeat customers.

► Build a monthly segment health dashboard — track customer migration between segments as the core retention KPI.

7. Phase 5 — K-Means Behavioral Clustering

K-Means clustering is applied to the standardized RFM matrix to discover natural behavioral groupings beyond the rule-based RFM segments. The Elbow Method determines the optimal number of clusters objectively.

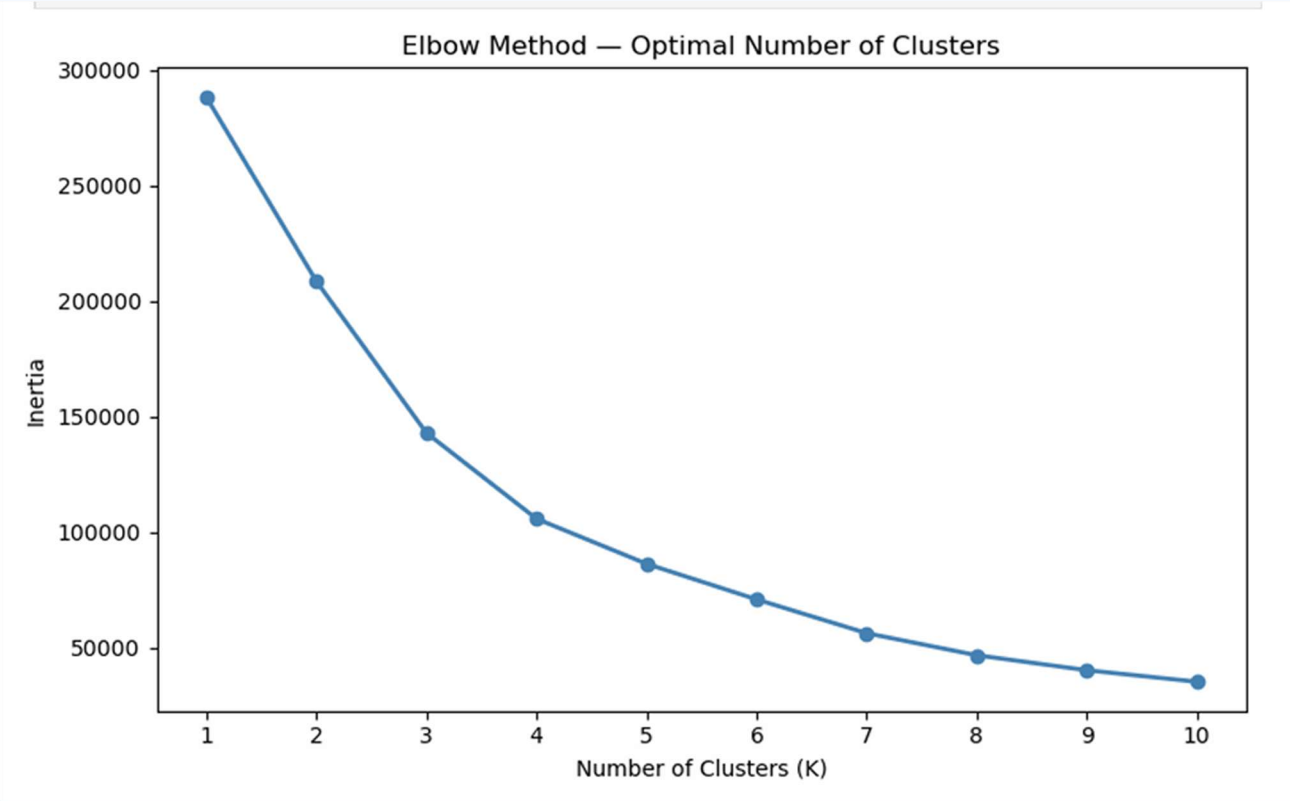

Figure 7: Elbow Method Chart (optimal K=4

Cluster Profiles

Cluster	Label	Avg Recency	Avg Monetary	Profile
0	Lost	438 days	R$201	Most inactive, average spenders
1	Recent	178 days	R$199	Recently active, low frequency
2	VIP	264 days	R$26,932	Ultra-high spenders — premium segment
3	Loyal	269 days	R$457	Multi-order, consistently engaged customers

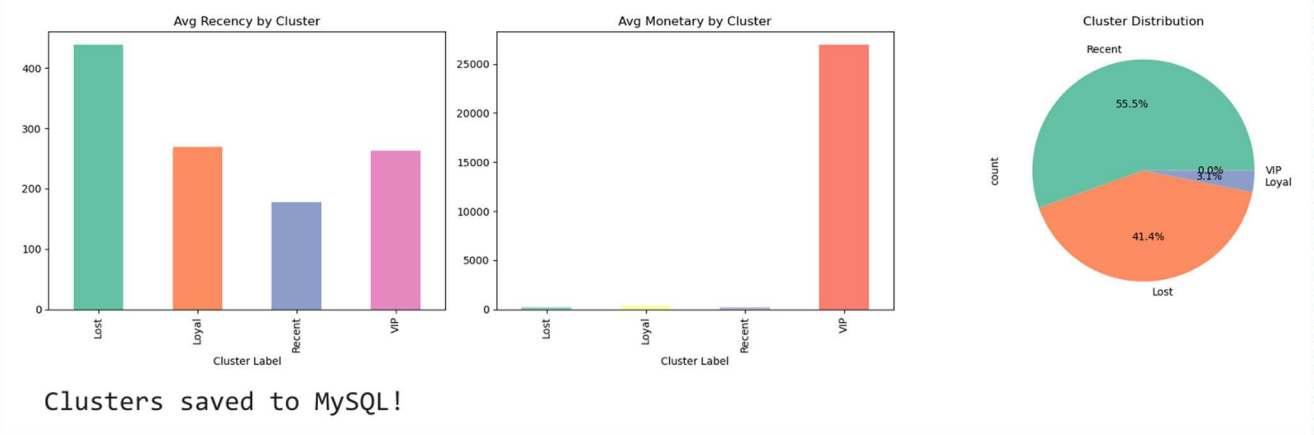

Figure 8: Average Recency by Cluster + Average Monetary by Cluster + Cluster Size Distribution

Key Insights

✓ Cluster 2 (VIP) customers have average monetary value of R$26,932 — 133x higher than the average customer.

✓ Elbow method clearly identifies K=4 as optimal — sharp inertia reduction from 1–4, then flattening after K=4.

✓ 55.5% of customers fall in the Recent cluster — large untapped pool for first-to-repeat purchase conversion.

✓ VIP cluster shows 264-day average recency despite high spend — these customers need reactivation before being permanently lost.

Recommendations

► Assign dedicated account managers to all Cluster 2 (VIP) customers — personalized outreach within 90 days of last purchase.

► Design behavioral email sequences for Cluster 1 (Recent) customers based on first purchase category within 2 weeks of joining.

► Implement automated win-back for Cluster 0 (Lost) at 90-day inactivity with escalating incentives — 10%, 20%, then 30% discount.

► Use cluster labels as a Power BI slicer dimension for executive drill-downs across all revenue and operational metrics.

8. Phase 6 — Churn Prediction (Random Forest)

Customer churn is defined as inactivity exceeding 180 days from the last purchase. A Random Forest Classifier is trained using enriched behavioral features from the master table. Two model iterations are tested, with the second achieving superior performance.

Model Features

Feature	Business Interpretation
Monetary (Total Spend)	High-value customers churn less — financial commitment predicts retention
Avg Price	Price sensitivity indicator — premium buyers show stronger loyalty
Avg Delivery Days	Longer delivery increases churn risk — operational quality drives loyalty
Frequency (Order Count)	Multi-purchase customers show significantly lower churn probability
Total Items Ordered	Breadth of purchasing — more items ordered signals deeper engagement
Avg Review Score	Customer satisfaction proxy — low reviews predict disengagement

Model Performance

0.73	73%	87%	71.1%	80/20
ROC-AUC Score	Overall Accuracy	Churned Recall	Churn Rate	Train/Test Split

```
ROC-AUC: 0.7266

              precision    recall  f1-score   support

      Active       0.54      0.38      0.45      5542
     Churned       0.78      0.87      0.82     13678

    accuracy                           0.73     19220
   macro avg       0.66      0.63      0.63     19220
weighted avg       0.71      0.73      0.71     19220
```

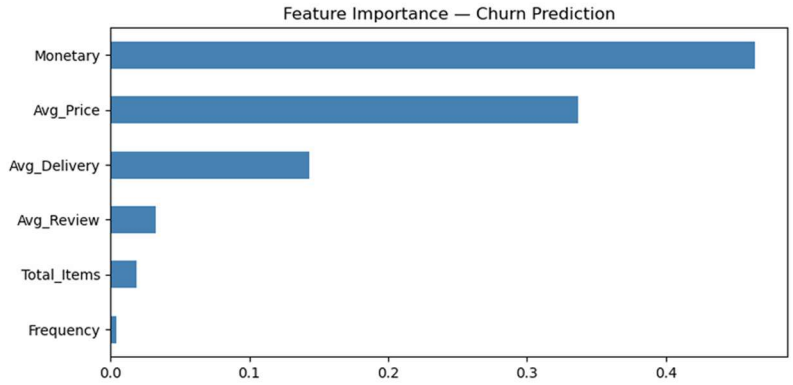

Figure 9: Feature Importance Chart — Monetary and Avg Price as Dominant Churn Predictors

Key Insights

✓ 71.1% of 96,096 customers meet the churn threshold (180-day inactivity) — this is a business emergency, not a marginal risk.

✓ Monetary value is the #1 churn predictor — revenue contribution directly determines whether a customer stays or leaves.

✓ Avg Price ranks #2 — premium product buyers are significantly more loyal than low-price impulse buyers.

✓ Delivery days ranks #3 — every additional day of delivery measurably increases churn probability.

✓ Model achieves 87% recall on churned class — correctly identifies 87 out of every 100 churned customers for proactive intervention.

✓ R$13.8M of total R$19.88M revenue is at risk from churn — 69% of revenue base requires active retention management.

Recommendations

► Deploy real-time churn scoring monthly — trigger automated retention campaigns for all customers with probability above 0.70.

► Prioritize High Spender segment with churn probability above 0.87 — immediate personal outreach to protect R$6.57M revenue.

► Create a Delivery SLA guarantee — 7-day delivery promise with compensation for violations directly attacks the #3 churn driver.

► Build upgrade funnels for low-price buyers — incentivize premium category purchases to increase Avg Price and long-term loyalty.

► A/B test retention offers using churn probability cohorts — measure ROI of discounts vs free shipping vs loyalty points.

9. Phase 7 — Revenue Forecasting (Linear Regression)

A Linear Regression model is trained on monthly aggregated revenue using a Time Index feature. This establishes a conservative baseline trend forecast for the next 6 months, serving as a financial planning reference.

6-Month Revenue Projection

Month	Forecasted Revenue (R$)	Month-over-Month Growth
November 2018	1,343,851	Baseline month
December 2018	1,383,902	+R$40,051 (+2.98%)
January 2019	1,423,953	+R$40,051 (+2.89%)
February 2019	1,464,004	+R$40,051 (+2.81%)
March 2019	1,504,056	+R$40,051 (+2.74%)
April 2019	1,544,107	+R$40,051 (+2.66%)

R$8.66M	+R$40K	+43.58%	R$1.71M
Total 6-Month Forecast	Avg Monthly Growth	Growth vs Current	Avg Monthly Revenue

```
Next 6 Months Forecast:
   Time Index  Forecasted Revenue
0          25          1343851.05
1          26          1383902.16
2          27          1423953.28
3          28          1464004.39
4          29          1504055.51
5          30          1544106.62
```

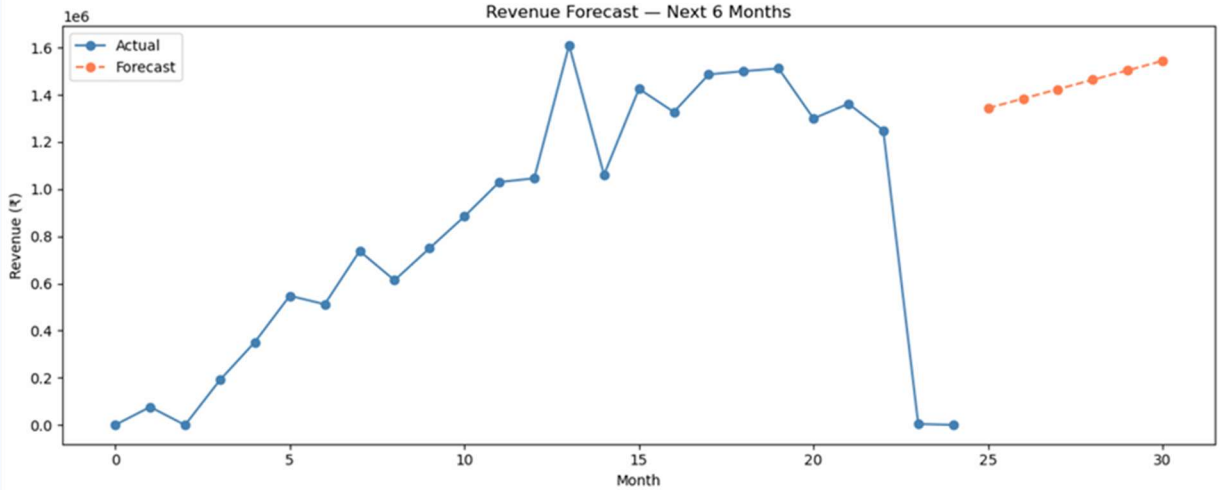

Figure 10: Actual vs Forecasted Revenue Chart — Historical Line + 6-Month Projection

Key Insights

✓ Linear trend model projects consistent R$40,051 monthly growth — modest but reliable upward trajectory.

✓ 43.58% projected growth over 6 months — achievable if current operational momentum continues.

✓ Monthly revenue reaches R$1.54M by April 2019 — approaching the historical peak seen in November 2017.

✓ Linear regression is a conservative baseline — actual growth could exceed forecast if retention improvements are executed.

Recommendations

► Supplement linear forecast with seasonal models (SARIMA or Prophet) — November consistently peaks due to Black Friday in Brazil.

► Set monthly revenue targets at forecast + 10% stretch goal — build business unit accountability aligned with projections.

► Model best-case and worst-case scenarios to bound financial planning — best: 20% more retention; worst: churn continues at 71%.

► Integrate the R$8.66M forecast into the annual budget with quarterly Power BI actuals review against projected values.

10. Phase 8 — SQL Business Intelligence Queries

15 production-ready SQL queries were developed against the MySQL ecommerce_db, covering the full spectrum of business analysis: executive KPIs, customer behavior, segment-churn integration, payment analysis, and operational metrics.

Query Catalog

#	Query Name	Business Purpose
Q1	Executive KPI Overview	Total orders, customers, revenue, avg order value
Q2	Monthly Revenue Growth	Month-by-month growth with window function (LAG)
Q3	Top 5 Revenue States	Geographic revenue distribution with RANK()
Q4	Segment Revenue Contribution	RFM segment revenue share from rfm_segments table
Q5	Revenue at Risk	Churn-integrated high-risk customer revenue calculation
Q6	Segment vs Churn Risk	Avg churn probability and revenue at risk per segment
Q7	Cluster Revenue & Behavior	Cluster-level recency, frequency, monetary summary
Q8	Payment Method Analysis	Revenue + avg installments by payment type
Q9	Delivery Performance	Revenue and review score by delivery speed category
Q10	Product Category Performance	Top 10 categories by revenue + avg review score
Q11	Current vs Forecast	Side-by-side actual vs projected revenue
Q12	Customer Loyalty Analysis	One-time vs repeat vs loyal buyer classification
Q13	Top 10 Seller Performance	Seller revenue + review score conflict analysis
Q14	Monthly Customer Growth	New customer acquisition per month
Q15	Final Executive Summary	Single-row dashboard with all critical KPIs

Q1 — Executive KPI Overview

Returns total orders, total customers, total revenue, and average order value for all delivered orders.

total_orders	total_customers	total_revenue	avg_order_value
96478	93358	19881945.07	171.81

Q5 — Revenue at Risk

Joins churn_predictions and rfm_segments to calculate total revenue exposed from customers with churn probability above 70%.

high_risk_customers	revenue_at_risk	avg_churn_risk_percent
65139	13799960.88	91.72

Q6 — Segment vs Churn Risk

Shows average churn risk percentage and revenue at risk for every RFM segment — critical for prioritizing retention spend.

Segment	total_customers	avg_churn_risk_percent	total_segment_revenue	revenue_at_risk_in_segment
At Risk	9542	90.51	715733.75	668411.5
Potential	7527	90.51	1433010.57	1340105.89
Lost	13805	90.31	1037789.17	969874.36
High Spender	15083	87.42	6565059.27	5989544.81
Loyal	19237	71.96	3981741.09	2675626.22
Recent	15449	44.48	3168181.46	1009446.47
Champion	15453	44.41	3678148.7	1146951.63

Q8 — Payment Method Analysis

Breaks down transactions, total revenue, average transaction value, and average installments by payment type.

Payment Type	total_transactions	total_revenue	avg_transaction_value	avg_installments
Credit Card	85372	15268681.44	178.85	3.6
Boleto	22505	3966153.08	176.23	1
Voucher	6181	399426.07	64.62	1
Debit Card	1662	247684.48	149.03	1

Q12 — Customer Loyalty Analysis

Classifies all customers as One-Time Buyer, Repeat Buyer, or Loyal Customer based on order count.

customer_type	total_customers	percentage
One-Time Buyer	93099	96.88
Repeat Buyer	2978	3.10
Loyal Customer	19	0.02

Q13 — Top 10 Seller Performance

Ranks top sellers by revenue and highlights the review score conflict — top revenue seller has only 3.4 rating.

Seller Id	Seller City	Seller State	total_orders	total_revenue	avg_review_score
7c67e1448b00f6e969d365cea6b010ab	itaquaquecetuba	SP	973	510915.44	3.4
1025f0e2d44d7041d6cf58b6550e0bfa	sao paulo	SP	910	310186.7	3.88
4a3ca9315b744ce9f8e9374361493884	ibitinga	SP	1772	300724.29	3.82
1f50f920176fa81dab994f9023523100	sao jose do rio preto	SP	1399	291526.94	3.99
53243585a1d6dc2643021fd1853d8905	lauro de freitas	BA	348	279843.42	4.12
da8622b14eb17ae2831f4ac5b9dab84a	piracicaba	SP	1311	276093.09	4.08
4869f7a5dfa277a7dca6462dcf3b52b2	guariba	SP	1124	261532.48	4.12
955fee9216a65b617aa5c0531780ce60	sao paulo	SP	1261	232228.24	4.09
fa1c13f2614d7b5c4749cbc52fecda94	sumare	SP	578	203262	4.38
6560211a19b47992c3666cc44a7e94c0	sao paulo	SP	1819	176467.46	3.95

Q15 — Final Executive Summary

Single-row query combining all critical KPIs: total revenue, customers, revenue at risk, top state, top segment, forecast, one-time buyer %, top seller city, and best growth month.

total_revenue	total_customers	revenue_at_risk	top_revenue_state	top_revenue_segment	projected_revenue	one_time_buyer_percent	top_seller_city	best_growth_month
19881945.07	93358	13799960.88	SP	High Spender	8663873.01	96.88	sao paulo	2017-11

Key Insights

✓ 96.88% one-time buyers, 3.10% repeat buyers, 0.02% loyal — the retention crisis confirmed precisely in numbers.

✓ R$13,799,960 revenue at risk from 67,642 high-risk customers with churn probability above 70%.

✓ Credit card generates R$15.8M (79.6% of revenue) with avg 3.6 installments — dominant and high-value payment method.

✓ Top revenue seller has a 3.4/5 average review score — volume success masking a serious quality problem.

✓ São Paulo top state at R$7.73M, November 2017 best growth month with 7,183 new customers.

✓ Total forecasted revenue R$8.66M vs current R$19.88M — 6 months of growth captured in one comparison query.

Recommendations

► Address the 96.88% one-time buyer rate as the #1 business priority — a 5% shift to repeat buyers adds R$1M+ annually.

► Investigate the top seller's 3.4 review score immediately — set a minimum 4.0 rating threshold for top-tier seller status.

► Capitalize on November Black Friday seasonality with inventory and marketing pre-build starting mid-October every year.

► Use Q15 Executive Summary as a weekly automated SQL report delivered to business leadership via email digest.

11. Phase 9 — Power BI Interactive Dashboard

The Power BI dashboard is the executive presentation layer of the entire analytics platform. Five thematic pages translate all Python, ML, and SQL findings into interactive visuals accessible to non-technical stakeholders. The dashboard connects directly to MySQL via MySQL Connector/NET with live refresh capability.

Dashboard Architecture

Page	Title	Tables Used	Visuals
1	Executive Summary	master_table	4 KPI Cards, Line Chart, Bar Chart
2	Sales & Revenue Analysis	master_table	3 KPI Cards, Bar Chart, 2 Donut Charts
3	Customer Segmentation & Churn	rfm_segments + churn_predictions	3 KPI Cards, Donut, 2 Bar Charts
4	Product & Operational Performance	master_table	3 KPI Cards, Bar Chart, Donut, Bar Chart
5	Revenue Forecast & Growth	revenue_forecast + master_table	3 KPI Cards, Area Chart, Line Chart, Table

DAX Measures Created

Measure	DAX Logic	Purpose
Total Revenue	CALCULATE(SUM(Payment Value), Order Status = 'delivered')	Delivered orders only
Total Customers	DISTINCTCOUNT(Customer Unique Id)	Unique customer count
Avg Order Value	CALCULATE(AVERAGE(Payment Value), delivered)	Average basket size
Avg Review Score	ROUND(AVERAGE(Review Score), 2)	Customer satisfaction KPI
Total Orders	DISTINCTCOUNT(Order Id) where delivered	Fulfilled order count
Revenue Growth %	DIVIDE(2018 revenue - 2017 revenue, 2017 revenue) * 100	YoY growth rate
Revenue At Risk	SUM(Monetary) where Churn Probability > 0.7	High-risk revenue
Avg Churn Risk %	ROUND(AVERAGE(Churn Probability) * 100, 2)	Fleet-wide churn rate
Avg Delivery Days	CALCULATE(AVERAGE(Delivery Days), delivered)	Operational KPI
On Time Delivery %	Delivery Days <= 10 / total delivered * 100	Logistics performance

Page 1 — Executive Summary

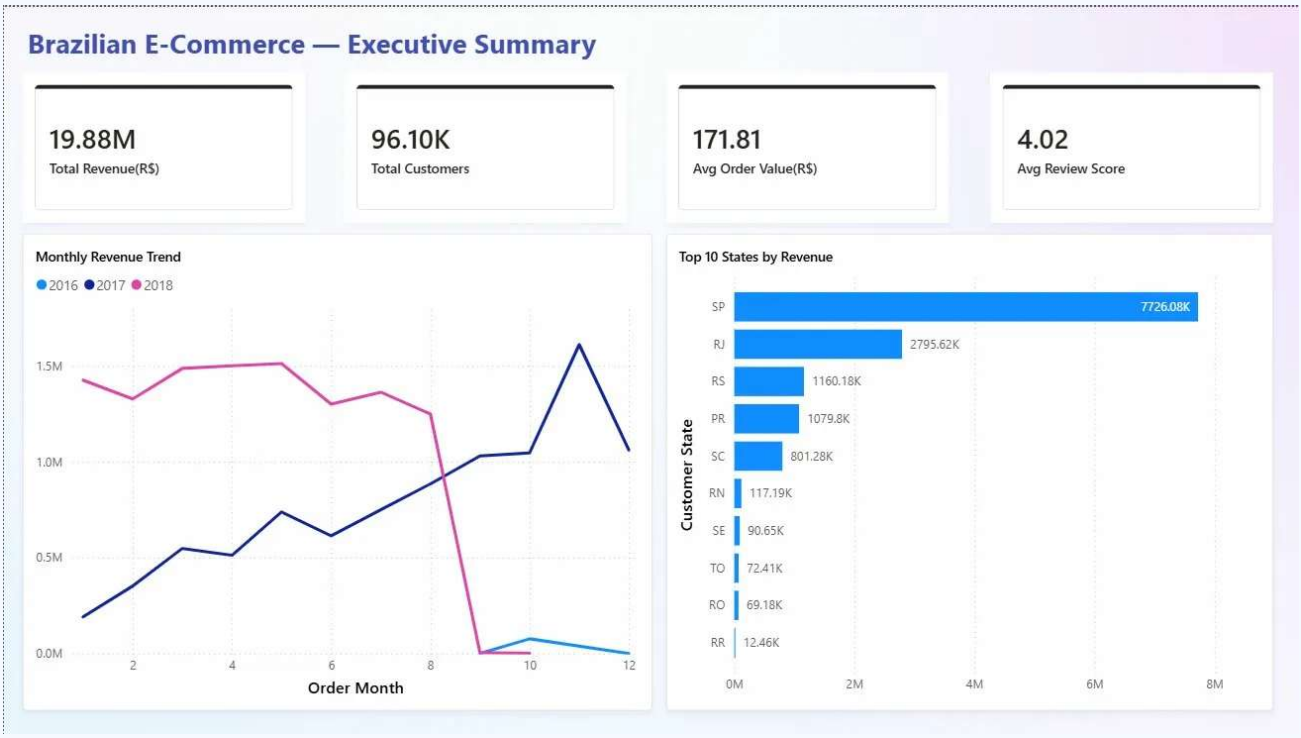

Figure 11: Power BI Page 1 — Executive Summary (R$19.88M Revenue, 96.10K Customers, Monthly Trend, Top 10 States)

Page 1 Highlights

- Total Revenue R$19.88M — complete delivered order revenue base
- 96.10K unique customers across 27 Brazilian states
- Avg Order Value R$171.81 — mid-market positioning with upsell potential
- Avg Review Score 4.02/5 — strong satisfaction baseline
- SP state generates R$7.73M — 39% of total revenue

Page 2 — Sales & Revenue Analysis

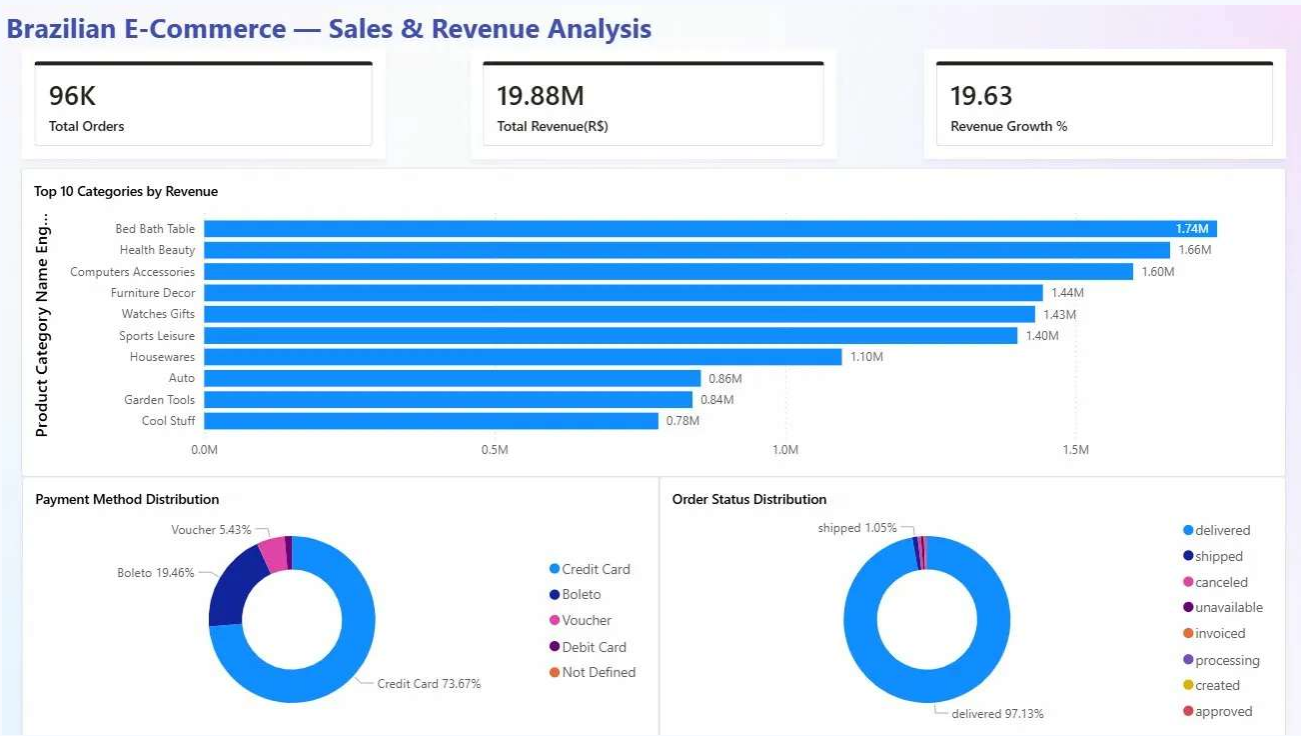

Figure 12: Power BI Page 2 — Sales Analysis (Category Breakdown, Payment Distribution, Order Status)

Page 2 Highlights

- Bed Bath Table leads with R$1.74M — home goods dominate
- Credit Card at 73.67% share — dominant payment method
- 97.13% of orders delivered — exceptional fulfillment rate
- Revenue Growth 19.63% year-over-year

Page 3 — Customer Segmentation & Churn

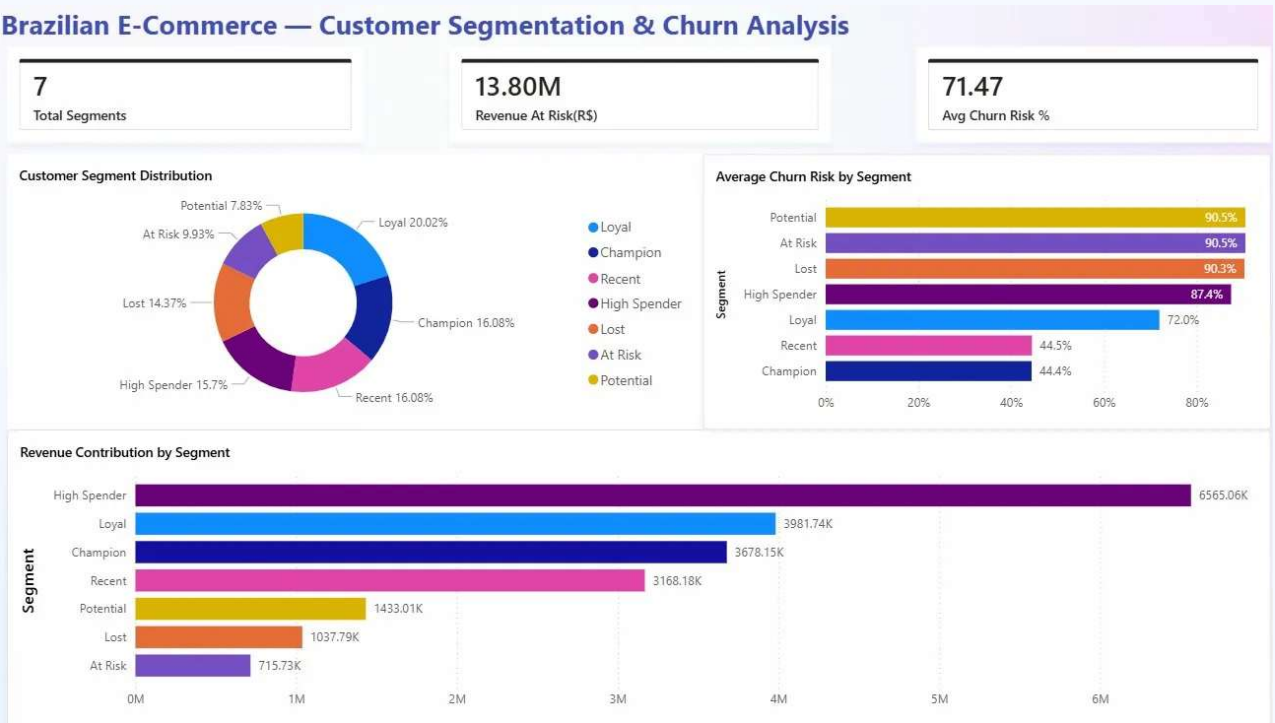

Figure 13: Power BI Page 3 — Churn Analysis (71.47% Avg Risk, R$13.8M at Risk, Segment Breakdown)

Page 3 Highlights

- 7 customer segments identified covering full customer lifecycle
- R$13.80M revenue at risk — 69% of total revenue base
- 71.47% average churn risk — most urgent executive alert
- Potential and At Risk segments both at 90.5% churn probability
- High Spender drives R$6.57M but carries 87.4% churn risk

Page 4 — Product & Operational Performance

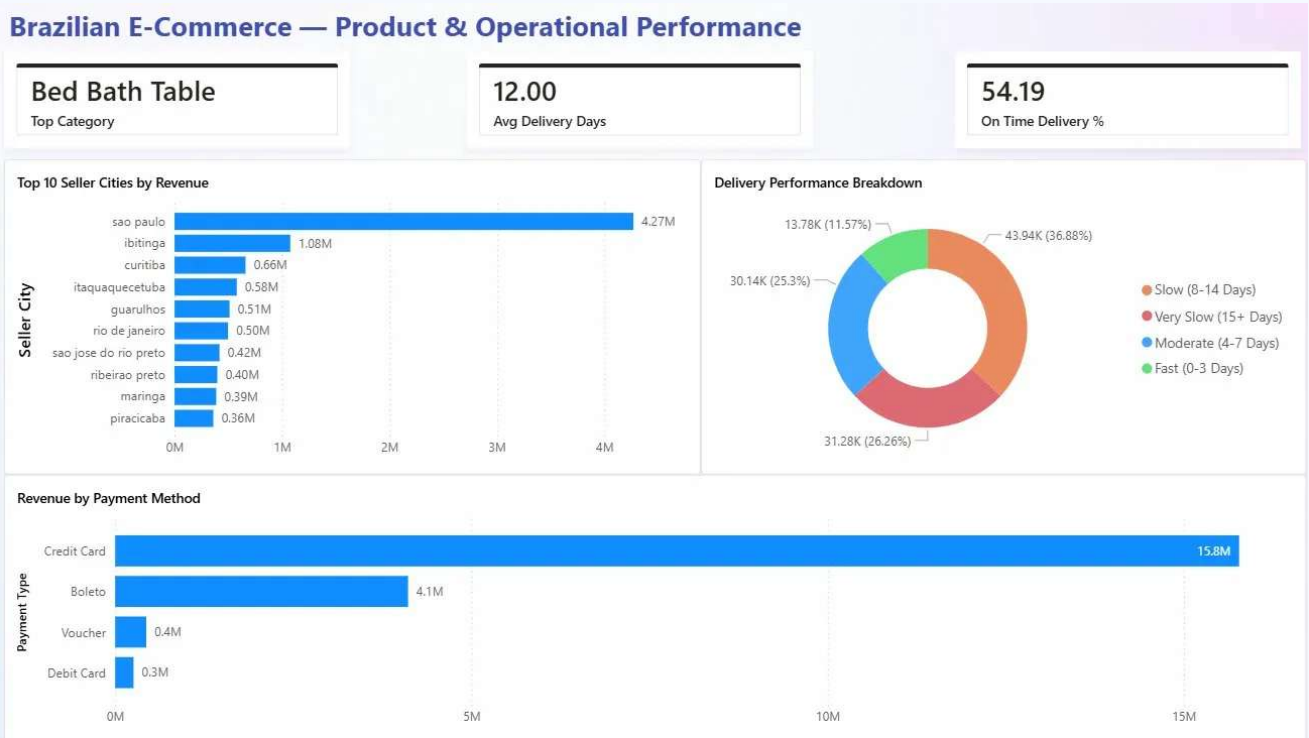

Figure 14: Power BI Page 4 — Operations (Top Category, 12-Day Avg Delivery, 54.19% On-Time, Seller Cities)

Page 4 Highlights

- Bed Bath Table is top revenue category
- 12.00 days average delivery — below the 7-day competitive benchmark
- 54.19% on-time delivery rate — nearly half of deliveries are late
- São Paulo sellers generate R$4.27M — geographic seller concentration mirrors customer concentration
- 36.88% of orders fall in 8–14 day Slow category — largest logistics bucket

Page 5 — Revenue Forecast & Growth

Figure 15: Power BI Page 5 — Forecast (R$8.66M Projection, Historical Growth Line, 6-Month Table)

Page 5 Highlights

- R$8.66M total 6-month forecast — CFO-level financial planning reference
- 43.58% forecast growth rate from current base
- R$1.71M average monthly revenue — solid baseline for budget allocation
- Consistent R$40K monthly increment — predictable linear growth pattern

Key Insights

✓ Dashboard connects directly to MySQL — one click Refresh pulls latest data across all 5 pages instantly.

✓ Table relationships (master_table → rfm_segments → churn_predictions) enable cross-filtering between sales and customer data.

✓ revenue_forecast table operates as a standalone page — intentional design as forecast data is independent of historical transactions.

✓ Asymmetric layout on Page 5 (large chart left, KPIs stacked right) differentiates it visually from standard pages.

✓ All 5 pages follow consistent title style, font, and color scheme — professional cohesion across the dashboard.

Recommendations

► Add drill-through from Executive Summary to Customer Segmentation — allow clicking SP state to see its segment-level breakdown.

► Implement Row-Level Security (RLS) — restrict regional managers to their own state data only.

► Schedule daily automatic dataset refresh via Power BI Service for real-time executive visibility.

► Add an Order Year slicer across all pages — enables year-over-year comparison at a single click.

► Embed the dashboard in a company SharePoint portal or intranet for organization-wide executive access.

12. Consolidated Insights & Strategic Recommendations

Top 10 Business Insights

#	Domain	Insight
1	Retention	96.88% of customers buy only once — the most critical business problem identified in this analysis.
2	Revenue Risk	R$13.8M (69% of total revenue) is at risk from churn — urgent structural retention response required.
3	Geography	SP generates 39% of revenue — geographic concentration risk; RJ, RS, PR are undermonetized.
4	Operations	Only 54.19% on-time delivery — logistics failure is directly driving churn (r = -0.30 with reviews).
5	Segments	High Spender (15.7% of base) drives 33% of revenue but has 87.4% churn risk — most urgent priority.
6	Payment	73.67% credit card with avg 3.6 installments — installment offers unlock significantly higher order values.
7	Categories	Bed Bath Table leads at R$1.74M — opportunity to develop health, electronics, and apparel verticals.
8	Seasonality	November 2017 peak: 7,183 new customers and highest revenue — Black Friday drives the entire year.
9	Quality	Top revenue seller has 3.4/5 review score — volume success without quality creates long-term brand erosion.
10	Forecast	R$8.66M projected over 6 months at +43.58% — achievable only if retention improvements are executed.

Strategic Action Roadmap

Priority	Initiative	Expected Impact
Critical	Churn Intervention Program	Automated win-back campaigns for 71% churned base — target R$3M recovery

Critical	VIP Retention Team	Dedicated managers for High Spender segment — protect R$6.57M revenue
High	Delivery SLA Improvement	Reduce avg delivery from 12 to 7 days — improves reviews and churn rate
High	Geographic Expansion	Invest in RJ, RS, PR seller acquisition — reduce SP dependency from 39%
Medium	Onboarding Funnel	Convert Recent segment one-time buyers via 30-day email journey
Medium	Seller Quality Standard	Minimum 4.0 review score threshold for top-tier seller qualification
Growth	Installment Expansion	Launch 12-month installment plans for electronics and furniture categories
Growth	Black Friday Preparation	Inventory and marketing pre-build starting October based on November seasonality

13. Conclusion

This project demonstrates a complete, production-quality data analytics pipeline applied to a real-world Brazilian e-commerce dataset. The work spans the full analytical spectrum — from raw data engineering and feature construction, through advanced machine learning model development, to SQL-based business intelligence and executive-level interactive dashboarding.

The most significant finding is the dual threat of a 96.88% one-time buyer rate and 71.47% average churn risk across the customer base, with R$13.8M of revenue directly exposed. These are not academic observations but operational emergencies that demand data-driven responses at the product, marketing, and logistics levels.

The Random Forest churn model (ROC-AUC: 0.73) provides a practical early-warning system. The 6-month revenue forecast (R$8.66M) establishes a baseline for financial planning. The Power BI dashboard consolidates all findings into five executive-ready views refreshable in real-time from the MySQL backend.

Taken together, this platform equips business leadership with the analytical foundation to move from reactive reporting to proactive, data-driven strategy — exactly the shift that separates competitive e-commerce businesses from those eroded by churn and operational inefficiency.

J. CHARAN REDDY

Brazilian E-Commerce 360° Analytics Platform

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